# Cognitive impairment and dementia in neurocysticercosis: a cross-sectional controlled study.

OBJECTIVES: Neurocysticercosis (NCYST) is the most frequent CNS parasitic disease worldwide, affecting more than 50 million people. However, some of its clinical findings, such as cognitive impairment and dementia, remain poorly characterized, with no controlled studies conducted so far. We investigated the frequency and the clinical profile of cognitive impairment and dementia in a sample of patients with NCYST in comparison with cognitively healthy controls (HC) and patients with cryptogenic epilepsy (CE). METHODS: Forty treatment-naive patients with NCYST, aged 39.25 +/- 10.50 years and fulfilling absolute criteria for definitive active NCYST on MRI, were submitted to a comprehensive cognitive and functional evaluation and were compared with 49 HC and 28 patients with CE of similar age, educational level, and seizure frequency. RESULTS: Patients with NCYST displayed significant impairment in executive functions, verbal and nonverbal memory, constructive praxis, and verbal fluency when compared with HC ($p < 0.05$). Dementia was diagnosed in 12.5% patients with NCYST according to DSM-IV criteria. When compared with patients with CE, patients with NCYST presented altered working and episodic verbal memory, executive functions, naming, verbal fluency, constructive praxis, and visual-spatial orientation. No correlation emerged between cognitive scores and number, localization, or type of NCYST lesions on MRI. CONCLUSIONS: Cognitive impairment was ubiquitous in this sample of patients with active neurocysticercosis (NCYST). Antiepileptic drug use and seizure frequency could not account for these features. Dementia was present in a significant proportion of patients. These data broaden our knowledge on the clinical presentations of NCYST and its impact in world public health.